INCH-POUND

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18 June 1993
SUPERSEDING
WS 23147D
26 November 1991

MILITARY SPECIFICATION

EXPLOSIVE, PLASTIC-BONDED, CAST PBXN-109

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of one type of cast plastic-bonded explosive, PBXN-109, for use in ordnance.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

P-D-680	Dry Cleaning and Degreasing Solvent

MILITARY

MIL-R-398	RDX

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 1376

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DOD-D-23443	Di-(2-ethylhexyl) Adipate, Technical
DOD-D-82727	Dibutyltin Dilaurate
MIL-T-82825	Triphenylbismuth

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-286	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-650	Explosive: Sampling, Inspection and Testing
MIL-STD-2073-1	DOD Materiel Procedures for Development and Application of Packaging Requirements
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (CAGE CODE 10001)

WS 16305	Isophorone Diisocyanate
WS 23148	Polybutadiene, Liquid, Hydroxyl-Terminated
WS 23149	N, N 2-Hydroxyethyl Dimethyl Hydantoin (Polyurethane Grade)
WS 23150	Aluminum Powder, Spherical
WS 23151	T-butylphenol Type Antioxidant

(Application for copies of OSs and WSs should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, 101 Strauss Avenue, Attn: Code 8410P, Indian Head, MD 20640-5035.)

CODE OF FEDERAL REGULATIONS (CFR)

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49 CFR 100-199 Transportation

(Copies of CFRs are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

CHEMICAL PROPULSION INFORMATION AGENCY (CPIA)

CPIA Publication No. 21 JANNAF Solid Propellant Mechanical Behavior Manual (Supplement)

(Application for copies should be addressed to the Chemical Propulsion Information Agency, Applied Physics Laboratory, Johns Hopkins University, Johns Hopkins Road, Laurel, MD 20810.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.2), a first article sample of the explosive material shall be manufactured using the same formulation, process, and procedures proposed for the production of the material. This sample shall be tested as specified in 4.4 prior to production, to determine that the contractor prior to starting production, is capable of producing the explosive material in conformance with the requirements of this specification.

3.2 Materials. The PBXN-109 explosive shall be a plastic-bonded and castable explosive material and shall contain a plastic material which is used to polymerize with the other ingredients to make it more flexible.

3.3 Formulation (M101). The formulation of explosive PBXN-109 shall comply with the requirements specified in Table I.

3.3.1 Equivalent ratio. The weight percent of isophorone diisocyanate shall be calculated (see 6.6 for calculation) using an NCO/OH ratio between 1.0/1 to 1.1/1. The NCO/OH ratios will be calculated using the equivalent weights of materials obtained by chemical analysis using the techniques of WS 23149 paragraph 4.5 (Quality conformance inspection), WS 16305 paragraph 4.5.2 (Isocyanate content), and WS 23148 paragraph 4.6.2 (Hydroxyl content determination). To avoid shelf-life problems with component

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materials, this chemical analysis must be conducted at least every 6 months on lots of materials being used. The weight percent of each ingredient shall be to two significant figures.

TABLE I. PBXN-109 formulation.

Ingredients	Percent (by weight)			Specification
	Nominal	Minimum	Maximum	
Sym-Cyclotrimethylene trinitramine (RDX) [1]				
Type II Class 1	64.00	56.5	65.5	MIL-R-398
Type II Class 5	0.00	0.0	7.0	MIL-R-398
Aluminum Powder, Spherical, Type IV	20.00	18.00	22.00	WS 23150
Polybutadiene, Liquid, Hydroxyl-Terminated, Type II (PolyBD) [2]	7.346	7.0	7.5	WS 23148
Di (2-ethylhexyl) adipate (DOA)	7.346	7.0	7.5	DOD-D-23443
2,2'-methylenebis (4-methyl-6-tertiarybutyl Phenol) antioxidant	0.10	0.09	0.11	WS 23151
N, N 2-Hydroxyethyl dimethyl-hydantoin (DHE)[2]	0.26	0.25	0.27	WS 23149
Triphenylbismuth (TPB) [3] (Dibutyltin dilaurate (DBTL)) [4]	0.02	0.015	0.045	MIL-T-82825 (DOD-D-82727)
Isophorone Diisocyanate (IPDI) [2]	0.9465	--	--	WS 16305

1 A maximum of 7 percent Class 5 RDX may be used in the formulation to improve processing characteristics. Addition of Class 5 RDX shall be made at the expense of Class 1 RDX. The total RDX weight shall be between 62 and 66 percent, with a nominal weight of 64 percent.

2 Typical equivalent analyses of binder components used to establish nominal composition are as follows:
PolyBD = 0.82 milliequivalent weight per gram (meq/g) (hydroxyl value) or 1219.5 g/eq (equivalent weight),
IPDI = 113.47 meq/g (equivalent weight),
DHE = 500 mg KOH/g (hydroxyl number) or 112.2 g/eq (equivalent weight),
NCO/OH Ratio = 1.0/1 (nominal).

3 Triphenylbismuth (TPB) is the recommended catalyst. Recommended nominal value is 0.02%.

4 Dibutyltin dilaurate (DBTL) may be substituted for TPB using between 0.002% and 0.02%. However, TPB is the recommended catalyst.

3.4 Cured explosive requirements (M102). The cured explosive shall meet the requirements specified in Table II.

TABLE II. Cured explosive requirements.

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Property	Minimum	Maximum	Test Method
Density, gram per cubic centimeter (g/cc) (at 25 degrees Celsius (•C))	1.60	1.70	4.6.1
Stress (maximum) pounds per square inch (at 25•C)	60	--	4.6.2
Strain, maximum stress, % (at 25•C)	12	--	4.6.2
Hardness, Shore A (at 25•C) (30 seconds)	30	--	4.6.3
Vacuum stability (at 100•C) milliliter (mL) gas/g/48 hours	--	0.5	4.6.4
Average composition: % RDX % Aluminum powder, spherical % Binder	 62.0 18.0 14.0	 66.0 22.0 18.0	4.6.5

3.5 Workmanship. The explosive material shall be free from foreign materials and manufactured in a manner to ensure compliance with all of the requirements of this specification. The workmanship exhibited in the first article sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum requirement for all units offered for acceptance.

3.6 Classification of characteristics. Characteristics verified by the inspections and tests specified herein are classified as Critical, Major and Minor in accordance with DOD-STD-2101. Critical characteristics are identified by the symbol (C) and Major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristics. Characteristics not annotated by the classification code symbol are classified as Minor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified in the applicable paragraphs in this specification, inspections shall be performed in room ambient conditions.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on samples which have been produced with equipment and procedures normally used in production. Any production prior to acceptance of the first article sample is at the contractor's risk.

4.4.1 Sample for first article inspection. The sample shall consist of two sets of representative material selected from a minimum batch of 225 kilograms after isocyanate has been added. A set will contain approximately 2,000 g of explosive and shall be composed of a specimen of explosive selected from the first third of the batch and a specimen of explosive selected from the final third of the batch.

4.4.2 Inspection routine. Material from one of the two sample sets from the batch shall be subjected to the tests specified in 4.6. The second sample set shall be forwarded to the test activity specified in the contract or order (see 6.2) in a form suitable for the tests specified in 4.6.

4.4.3 Acceptance criteria. Failure to meet the requirements of this specification shall cause rejection of the first article sample.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the tests specified in 4.6.1, 4.6.3, and 4.6.5.

4.5.1 Sample for quality conformance. The sample shall be the same as that of 4.4.1 except that it shall consist of one set. Uncured material for analysis shall be selected from the same containers from which the cast samples are poured.

4.5.2 Sample identification. Each explosive sample container and specimen shall be identified with the following information:

a. Complete explosive designation.

b. Lot number.

c. Weight of the lot.

d. Manufacturer's name and plant designation.

e. Contract number.

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f. Date of manufacture of sample.

4.5.3 Acceptance criteria. Explosive samples which fail to meet any of the requirements of 3.4 when tested by the methods specified in 4.6 shall be classed as defective with rejection of the batch represented.

4.6 Test procedures. All explosive specimens, with the exception of samples used in the uncured PBXN-109 analytical procedures (4.6.5.3 and 4.6.5.4), shall be cured at 30 to 60•C, and shall pass the hardness requirements of Table II prior to subjecting them to the tests of 4.6.1 through 4.6.5. The uncured explosive samples should be cured after uncured compositional analysis.

4.6.1 Density test. Density shall be determined on two pie-shaped wedges cut from two separate cured test samples (one sample taken from the beginning of the pour and one sample taken from the end of the pour). Each sample shall weigh between 10 and 20 g. Density of the samples shall be determined on an analytical balance equipped for underwater weighing. The sample shall be weighed to an accuracy of 0.1 milligram (mg), first in air, then in water. Calculate the density according the following formula:

$$\Delta \quad \frac{\Omega\alpha}{\Omega\alpha \quad \Omega\beta} \cdot \Delta\omega$$

where:

D = Density of test sample, (g/cc)

Dw = Density of water at test temperature, (g/cc)

Wa = Weight of test sample in air, (g)

Wb = Weight of test sample in water, (g)

4.6.1.1 Density acceptance criteria. If one sample fails to meet the density requirements specified in Table II, another sample shall be taken from the same section of the mix and tested. If the second sample meets the density requirements, the batch shall be considered to meet the density requirements of this specification.

4.6.2 Stress and strain at maximum stress. Stress and strain at maximum stress shall be determined in accordance with JANNAF Solid Propellant Mechanical Behavior Manual, CPIA Publication No. 21, Section 4.3.2, Uniaxial Tensile Tests at Constant Strain Rate, November 1970. Class C testing shall be used, and final specimens may be die cut or milled from the half inch slices from the bulk sample. The specimens for testing may be selected from the first third of the batch or the final third when sampled as described in 4.4.1.

4.6.3 Shore A hardness test. The explosive of 4.4.1 or 4.5.1, in a suitable form as in 4.6.1, shall be tested for Shore A hardness. (Any sample which exhibits bleeding at the surface at the end of the cure period shall not be tested.) The hardness of the cured material shall be determined using a Type A-2 Durometer, with a maximum dial indicator or hand, together with a constant load operating stand (Code

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DRCL). Total weight of dead weight, holding screw, and durometer shall be 1065 g. The test specimen shall be placed under the durometer pointer with the pointer resting on the surface of the sample. Release the lock (called a "Pawl"). Do not permit the point to shock the sample. Record the instrument dial reading 30 seconds after release, using a stopwatch to measure the time. Determine three values each on the top and the bottom of the specimen. Record the average of the top average and the bottom average as the test value.

4.6.4 Vacuum stability test.

4.6.4.1 Capillary tube method. The capillary tube method of the vacuum stability test shall be carried out in a sealed glass unit (see Figure 1). These tubes shall be calibrated before use and shall contain a volume of approximately 10 mL. One millimeter (mm) (nominal) capillary bore Pyrex (or equivalent) tubing shall be used for the manometer. The vacuum stability test chamber is a constant temperature block that accurately maintains a temperature within ± 0.5•C.

a. **Calibration.** The volume of the hot zone shall be found by adding water from a burette to the open end of the sample tube until the water level reaches the height of the upper edge of the horizontal capillary tube (the point where the sample tube is to be sealed) allowing water to flow into the capillary tube to the 102 mm mark.

b. **Reference**. This volume of water in mL represents the volume of the hot zone for use in the calculations of 4.6.4.1e. Make a zero reference mark on the capillary tube 820 mm up from the bottom of the 920 mm vertical section. The apparatus shall be thoroughly dry before the test procedure is started.

c. **Preparation.** Introduce 4.0 mL of mercury into the cup at the base of the capillary tube and connect a vacuum line to its mouth. Evacuate the tube to approximately 1 mm of mercury. Remove the vacuum supply and allow the mercury to enter the capillary.

d. **Procedure.** Place the sample tube containing a 0.2 to 0.3 g specimen, weighed to the nearest 0.5 mg, into the stability block after an initial room temperature reading and barometer reading have been recorded. After the specimen has been in the block 1 hour, take a reading; then take a final reading 48 hours after the first reading. The difference in mm shall be corrected for changes in barometric pressure between the beginning and end of the test.

e. **Calculation.** All calculations shall be corrected to standard temperature and pressure (STP). Calculate the volume of gas evolved per gram of specimen as follows:

$$\varsigma \quad \frac{(\xi)\ (\psi)\ (\zeta)}{\omega}$$

where:

V = mL gas/gram specimen/48 hours at 100•C

x = Volume of hot zone (mL)

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y = Conversion factor = 0.000963 = $\frac{273}{373 \cdot 760}$

z = Corrected reading (mm)

w = Specimen weight (g)

This test shall be performed on two specimens from each lot. The volume of gas evolved per gram of specimen shall meet the vacuum stability requirements in Table II.

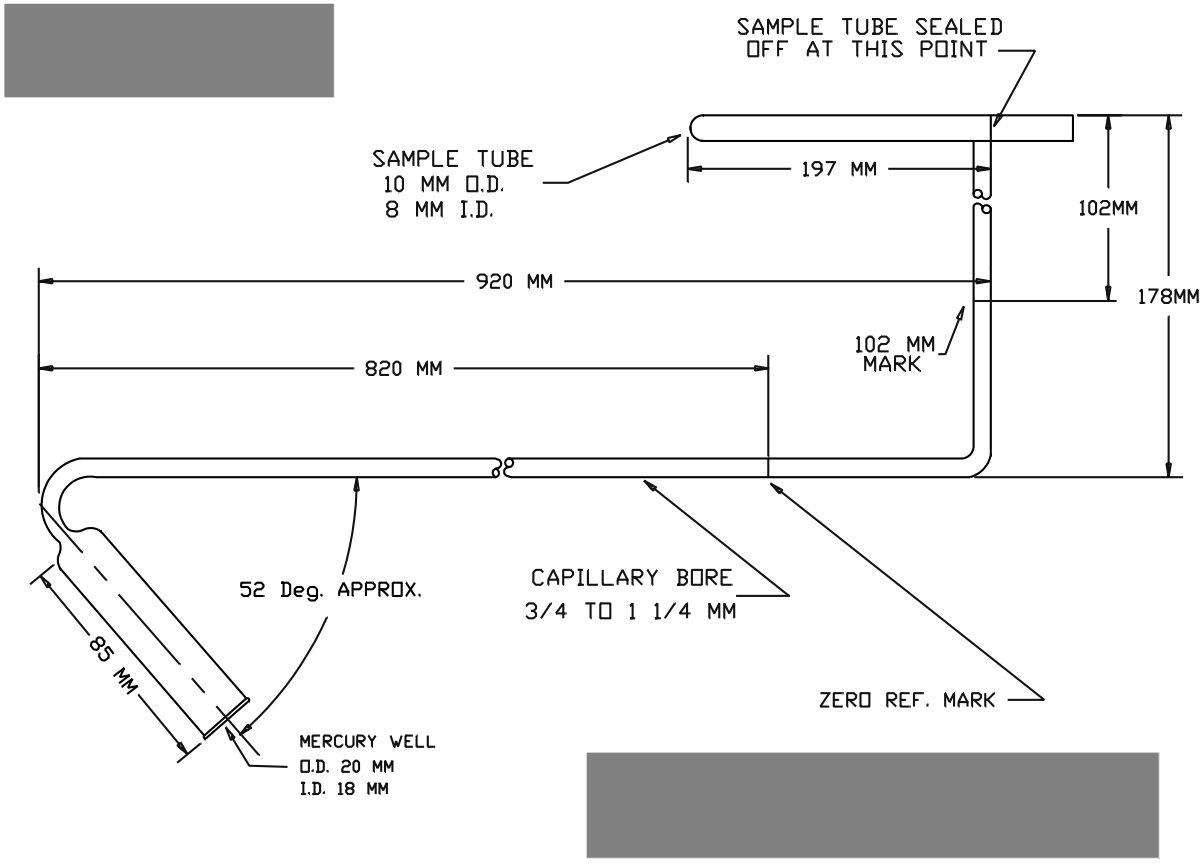


FIGURE 1. Vacuum stability test equipment.

4.6.4.2 Alternate method #1. An alternate method which may be used to determine vacuum stability shall be that described in MIL-STD-650, Method 503.1.1, except the sample size shall be 1 g; change Figure 1 horizontal capillary tube dimension to read either 26 or 15 centimeters (cm); correct the volume of the gas equation in accordance with 4.11 of MIL-STD-650, Method 503.1.1, except the actual equation shall be as follows:

$$\varsigma \quad [A \quad X \pm (B \quad H)] \frac{273\Pi}{760\ (273\ \tau)} \qquad [A \quad X \pm (B_i \quad H_i)] \frac{273\Pi_i}{760\ (273\ \tau_i)}$$

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where:

V = Volume of gas, (mL)

All other characteristics are identified in 4.11 of MIL-STD-650, Method 503.1.1. The sample shall be cut into small cubes with a maximum size of 3 mm before being placed in the vacuum stability test tube. Two samples, one from each section of the batch shall be used.

4.6.4.3 Alternate method #2. A second alternate method which may be used is MIL-STD-286, Method 403.1.2.

4.6.5 Composition analysis.

4.6.5.1 Cured PBXN-109, analytical procedure. Analyze, in duplicate, each sample of PBXN-109 received for analysis. This shall consist of a sample from the top and a sample from the bottom of the batch. Weigh a 0.5-g sample to the nearest 0.1 mg into a 100-mL beaker. Add 20 mL of acetone, boil and break up with a flat tipped stirring rod. Filter the solution through a weighed, medium porosity, sintered-glass crucible. Repeat the acetone extraction twice (total 60 mL acetone), then quantitatively transfer the residue to the crucible with acetone. Dry the crucible plus the residue at 100•C for 1 hour. Dilute the acetone solution to 100 mL in a 100-mL volumetric flask.

a. Weigh the crucible plus residue, then add 1 mL of 6N hydrochloric acid to the residue. Wait for the initial reaction with the aluminum to begin and subside, then add 5 mL more of the 6N hydrochloric acid. Let the reaction go overnight (about 15 hours). Dilute the acid with 10 mL of water and remove by suction. Rinse the resulting polymer at least 10 times with 10 mL of water, then with 10 mL acetone. Dry the crucible plus polymer at 100•C for 1 hour, cool, and weigh. The difference in weight after acetone extraction and after hydrochloric acid treatment is the weight of aluminum. The residue on the crucible after treatment with the hydrochloric acid is the polymer.

b. Weigh 0.5 g of di-n-butylphthalate into a 100-mL volumetric flask, dissolve in acetone, and dilute to volume with acetone. Mix 1 mL of this solution with 9 mL of the acetone solution from 4.6.5.1 above. Inject an aliquot of the mixture into a gas chromatography column containing SE-30 (2.5% on Chromosorb W HP 100/120 mesh, 4 feet x 1/4 inch) at 230•C (injection port 240•C; helium, 70 mL/minute) using flame ionization detector. Compare this chromatogram with a standard solution containing known amounts of di-n-butylphthalate and DOA.

c. Weigh 1.5 mg of TNT into a 10-mL volumetric flask, dissolve in methanol, and dilute to volume with methanol. Pipette 100 microliters of the acetone solution from 4.6.5.1 above into a 10-mL volumetric flask, and remove the acetone by heating in a boiling water bath. Add 1 mL of the TNT solution to the residue, dissolve by heating in the water bath, and dilute to volume with methanol. Prepare a standard methanol solution containing 6 parts per million (ppm) HMX, 30 ppm RDX, and 15 ppm TNT. Inject aliquots of the standard mixture and the unknown containing TNT onto an RP-8 HPLC column, elute with 40% methanol in water (2 mL/minute), and detect with 240 nanometers radiation.

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4.6.5.2 Alternate cured PBXN-109 analytical procedure. Analyze one top and one bottom specimen in duplicate. Cut the specimen into pieces measuring approximately 2 mm.

a. Weigh an approximate 0.5000-g specimen to the nearest 0.1 mg into a tared, medium porosity crucible with a glass stirring rod.

b. Extract the heptane-soluble materials using three 10 mL portions (approximate volumes) of n-heptane. Allow the n-heptane to soak for a few minutes, and mash the specimen using the glass stirring rod prior to filtering.

c. Dry the crucible plus residue at approximately 100•C until no n-heptane fumes remain. Cool the crucible plus residue to room temperature. Weigh the crucible, glass stirring rod, plus residue.

d. Calculate the percent n-heptane extractables as follows:

$$\text{Percent n-heptane extractables} = \frac{A - B}{X} \cdot 100$$

where:

A = Weight of crucible plus specimen and glass stirring rod, (g)

B = Weight of crucible plus residue and glass stirring rod after n-heptane extraction, (g)

C = Weight of specimen, (g)

e. Transfer the residue to a 100-mL or 150-mL beaker, and add approximately 20 mL of acetone. To extract the RDX, carefully boil the specimen and mash the pieces using a glass stirring rod. Filter the solution through the same crucible described in 4.6.5.2a above. Repeat the acetone extraction three times using approximately 20 mL of acetone each time.

f. Dry the crucible plus residue at approximately 100•C until no acetone fumes remain. Cool the crucible plus residue to room temperature. Weigh the crucible, glass stirring rod, plus residue. Calculate the percent RDX as follows:

$$\text{Percent RDX} = \frac{B - \Delta}{X} \cdot 100$$

where:

D = Weight of crucible plus residue and glass stirring rod after acetone extraction, (g)

g. Transfer the residue contained in the crucible to a 30-mL beaker. Add approximately 1 mL of 6N hydrochloric acid to the residue. Wait for the initial reaction with aluminum to begin and

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subside. Add approximately 5 mL of hydrochloric acid to the beaker. Place the beaker on a hot plate and warm it gently on low heat. Additional 6N hydrochloric acid may be added and the reaction allowed to continue until all the aluminum has been digested. The specimen may also be mashed using a glass stirring rod to help speed up aluminum digestion. Transfer the specimen to the same crucible described in 4.6.5.2a above, and filter the acid. Rinse the resulting polymer at least 10 times with 10 mL of distilled water. Finally, wash the sides of the crucible and polymer with 10 mL of acetone.

h. Dry the crucible plus residue at approximately 100•C for about 1 hour. Cool the crucible plus residue to room temperature. Weigh the crucible, glass stirring rod, plus residue. Calculate the percent aluminum as follows:

$$\text{Percent aluminum} \quad \frac{\Delta \quad E}{X} \cdot \quad 100$$

where:

E = Weight of crucible plus residue and glass stirring rod after acid extraction, (g)

i. The remaining residue in the crucible is the polymer. Calculate the percent polymer as follows:

$$\text{Percent polymer} \quad \frac{E \quad \Phi}{X} \cdot \quad 100$$

where:

F = Weight of crucible and glass stirring rod, (g)

j. Calculate the total percent binder as follows:

$$\text{Percent binder} \quad \text{percent n heptane extractables} \quad \text{percent polymer}$$

4.6.5.3 Uncured PBXN-109 analytical procedure. Analyze, in duplicate, each sample of PBXN-109 received for analysis. This shall consist of a sample from the top and a sample from the bottom of the batch.

a. Accurately weigh on an analytical balance to the nearest 0.1 mg an approximate 2-g sample of the uncured explosive into a tared, 30-mL fine porosity, sintered glass filtering crucible. Include the glass stirring rod in the tare weight of the crucible.

b. Extract at ambient temperature using a vacuum filtrator with 5 portions, 10 mL each, of the methylene chloride saturated with RDX. The extractions should have a stirring time of 1 minute each with the vacuum off while stirring.

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c. Dry the sample for 30 minutes between 90 and 100•C, cool and weigh.

d. Calculate the percent binder as follows:

$$\text{Percent binder} = \frac{A - B}{X} \cdot 100$$

where:

A = Weight of crucible plus sample, (g)

B = Weight of crucible plus residue, (g)

C = Weight of sample, (g)

e. Extract residue remaining on the crucible with 5 portions, 10 mL each, of the hot acetone (approximately 50•C) on a vacuum filtrator. Stir each extraction for 30 seconds with vacuum off.

f. Dry sample for 30 minutes between 90 and 100•C, cool and weigh.

g. Calculate the percent RDX as follows:

$$\text{Percent RDX} = \frac{D - E}{X} \cdot 100$$

where:

D = Weight of crucible plus residue before acetone extraction, (g)

E = Weight of crucible plus residue after acetone extraction, (g)

C = Weight of sample, (g)

h. Calculate the percent aluminum as follows:

$$\text{Percent aluminum} = 100\% - (\% \text{ binder} + \% \text{ RDX})$$

4.6.5.4 Alternate uncured PBXN-109 analytical procedure. In the following procedure, the solvent shall be either dry cleaning solvent, P-D-680, Type II, or n-heptane, reagent grade. Analyze one top and one bottom specimen in duplicate.

a. Weigh approximately 2 g, to the nearest 0.1 mg, of the specimen into a tared, fine porosity fritted glass crucible with a glass stirring rod.

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b. Extract the specimen with a total of approximately 80 mL of solvent. Extract with 3 portions of approximately 25 mL, stirring for about 15 seconds after the addition of each portion before applying vacuum. Rinse the contents and crucible with 5 mL of solvent while applying vacuum.

c. Place the crucible on a metal grate or rack that exposes the bottom of the crucible, and place the rack in a vacuum oven maintained at 85 ± 5•C for approximately 1 hour (or until all solvent has been removed) at 20 mm of mercury absolute pressure or less with a slight air bleed.

d. Remove the rack and crucible from the oven. Allow the crucible and contents to cool, and weigh them with the glass stirring rod.

e. The resultant loss in weight is binder.

f. Extract the crucible and contents with acetone, heated to approximately 50•C, following the procedure in 4.6.5.4b.

g. After the final rinse of acetone, allow the crucible to remain on the filtrator, under vacuum, while stirring the contents for approximately 1 minute, or until the presence of acetone is no longer detected. The crucible may be placed in an oven heated to 85•C for 10 minutes, if preferred.

h. Remove the crucible from the filtrator or oven. Weigh the crucible and contents with the glass stirring rod.

i. The resultant loss in weight is RDX, and the residue is aluminum.

4.6.5.5 Acceptance criteria for composition analysis. Record batch components to the nearest 0.1 percent as the average of the top and bottom samples. Failure of any composition shall require a repeat analysis on the backup sample. The backup sample shall be maintained under refrigeration. This will delay polymerization for a maximum of 24 hours.

4.7 Inspection of packaging. The packaging, packing, and marking shall be visually inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Packaging. The PBXN-109 explosive shall be packaged in accordance with 49 CFR 100-199 with Level C protection as defined in MIL-STD-2073-1.

5.2 Packing. The material packaged as specified in 5.1 shall be packed to ensure carrier acceptance and safe delivery to the destination at the lowest applicable rate. The protection shall be Level C as defined in MIL-STD-2073-1.

5.3 Marking.

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5.3.1 Special marking. Special explosive and precautionary container markings shall be in accordance with 49 CFR 100-199.

5.3.2 Normal marking. In addition to the special marking requirements specified in 5.3.1, all container marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. PBXN-109 is a high-energy explosive intended for use in ordnance. Safe use of PBXN-109 explosive is application dependent. When loading this explosive into a munition or test unit, the load design and materials that come in contact with the explosive must be closely examined to ensure that an unsafe condition, such as incompatibility, does not exist. The load design should be documented in a specification such as a Navy Munitions Data to ensure this safety.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Whether a first article sample is required (see 3.1).

d. The facility designated to perform inspection requirements if other than the contractor (see 4.1).

e. Activity to receive second first article sample (see 4.4.2).

k. Safety precaution requirements (see 6.4).

6.3 First article. When a first article sample is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Safety precautions. The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3.

NOTE: When this document is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.5 Definition.

6.5.1 Batch. A batch is that quantity of material which has been subjected to one or more chemical or physical processes (or combination thereof) intended to produce a desired product having substantially uniform characteristics.

6.6 Equivalent ratio. The following method is used to calculate the weight of IPDI required for a 1.0/1.0 NCO/OH ratio to react with the following materials:

a. For PolyBD (OH-terminated):

$$\Xi \quad \frac{(E\theta)\ (\Psi)\ (P)}{E\pi}$$

where:

X = Weight of IPDI required to react with PolyBD

Eq = Equivalent weight of IPDI, (g/eq)

Y = Weight of PolyBD (OH-terminated)

R = NCO/OH ratio (one for PBXN-109)

Ep = Equivalent weight of PolyBD (g/eq)

b. For Dantocol DHE:

$$\Xi \quad \frac{(E\theta)\ (O\eta\pi)\ (\Psi)\ (P)}{(56.1)\ (1000)}$$

where:

X = Weight of IPDI required to react with Dantocol

Eq = Equivalent weight of IPDI (gm/eq)

Ohp = Hydroxyl number of Dantocol DHE

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Y = Weight of Dantocol DHE

R = NCO/OH ratio (one for PBXN-109)

6.7 Subject term (key word) listing.

Composition analysis
DBTL
DHE
DOA
IPDI
PolyBD
Polymer
PBX
RDX
TPB
Vacuum stability test

Preparing Activity:
Navy - OS
(Project 1376-N459)

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NOTICE OF CANCELLATION	CAGE CODE 53711	INCH-POUND

WS 23147D
NOTICE 1
18 June 1993

MATERIAL SPECIFICATION

EXPLOSIVE, PLASTIC-BONDED, CAST PBXN-109

WS 23147D, 26 November 1991, is hereby canceled. Future acquisition for this material should refer to MIL-E-82886(OS).

(Copies of referenced federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Naval Surface Warfare Center, Dahlgren Division
White Oak Detachment (Code N741)
10901 New Hampshire Avenue
Silver Spring, MD 20903-5000

AMSC N/A **FSC 1375**
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